

Chapter 17: Program Management for AI

Learning Objectives

By the end of this chapter, you will:

- Understand how Program Management principles apply to enterprise AI initiatives.
 - Learn how AI TPMs manage scope, stakeholders, risks, timelines, and budgets for AI programs.
 - Develop practical frameworks for delivering AI products from ideation to production.
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Beginner Explanation

Projects deliver a single outcome.

Programs deliver multiple related outcomes.

For example:

Project	Program
Build an HR Bot	Enterprise AI Transformation
Deploy Azure OpenAI	AI Center of Excellence
Implement RAG	Digital Workplace Modernization

A Program Manager coordinates:

- People
- Processes
- Technology
- Risks
- Budgets
- Timelines

An AI TPM does the same—but for AI systems.

Responsibilities of an AI TPM

- Define scope.
- Create roadmaps.
- Track milestones.
- Manage risks.
- Coordinate stakeholders.
- Report status.
- Drive execution.

Think of an AI TPM as the conductor of an orchestra.

Without a conductor, every musician may be talented, but the performance will not be synchronized.

Intermediate Explanation

AI programs are inherently more complex than traditional software initiatives because they involve:

- Models
- Data
- Cloud services
- Governance
- Evaluations
- Compliance
- Continuous monitoring

AI Program Lifecycle

Phase	TPM Responsibility
Initiation	Define objectives
Planning	Create roadmap
Execution	Manage delivery
Monitoring	Track KPIs
Closure	Capture lessons learned

Core Program Management Areas

Area	Description
Scope	What will be delivered?
Schedule	When will it be delivered?
Budget	What will it cost?
Risks	What can go wrong?
Stakeholders	Who needs updates?
Quality	How is success measured?

AI TPM Success Metrics

- Delivery predictability
 - User adoption
 - Program ROI
 - Budget adherence
 - Risk reduction
 - Stakeholder satisfaction
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Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Ensures successful execution of AI programs across teams.
Risk	Proactively identifies and mitigates program risks.
Stakeholder Management	Maintains alignment across executives, engineering, and business teams.
Budget	Tracks spend and ensures programs remain financially viable.

Enterprise Example

Business Problem

A Fortune 500 organization launched an enterprise AI transformation initiative with the following goals:

- Deploy 10 AI use cases.
- Establish an AI Center of Excellence.
- Train 5,000 employees.
- Implement AI Governance.

The program involved:

- 15 teams
- 8 departments
- Multiple vendors
- Global stakeholders

Solution

Establish a centralized AI Program Management Office (AI PMO) led by AI TPMs.

Architecture

- Azure DevOps
- Azure OpenAI
- Azure AI Search
- Power BI
- Microsoft Teams
- Azure AI Foundry

Outcome

Metric	Result
Programs Delivered	10
Employee Adoption	87%

Metric	Result
Budget Variance	<5%
Stakeholder Satisfaction	94%

Lessons Learned

- Governance should be established early.
- Executive sponsorship is critical.
- Weekly reporting improves visibility.
- RAID management significantly reduces surprises.

Azure Implementation

Services Used

Service	Purpose
Azure AI Foundry	AI project management and evaluations
Azure OpenAI	AI application capabilities
Azure AI Search	Enterprise retrieval
Azure Functions	Workflow automation

Supporting Services

- Azure DevOps
- Power BI
- Microsoft Teams
- Azure Monitor

Implementation Workflow

Step 1: Define Program Objectives

Examples:

- Increase productivity by 20%.
- Reduce operational costs.
- Improve customer satisfaction.
- Accelerate AI adoption.

Step 2: Build Roadmap

Include:

- Milestones
 - Dependencies
 - Risks
 - Deliverables
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Step 3: Execute Program

Manage:

- Teams
 - Vendors
 - Budgets
 - Communication
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Step 4: Monitor KPIs

Track:

- Schedule adherence
 - Cost performance
 - User adoption
 - Delivery health
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Step 5: Conduct Retrospectives

Capture:

- Lessons learned
 - Improvements
 - Best practices
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Mermaid Diagram

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```

```
A[Business Vision]
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```
A --> B[AI Program]
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B --> C[Roadmap]
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C --> D[Delivery Teams]

D --> E[AI Solutions]

E --> F[Business Outcomes]

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## AI Program Governance Model

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graph LR
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A[Executive Sponsor]

A --> B[AI TPM]

B --> C[Engineering]

B --> D[Security]

B --> E[Product]

B --> F[Operations]

C --> G[Production Delivery]

D --> G

E --> G

F --> G

TPM Checklist

- ☐ Define program objectives.
 - ☐ Create delivery roadmap.
 - ☐ Establish governance.
 - ☐ Build RAID logs.
 - ☐ Track milestones.
 - ☐ Review budgets.
 - ☐ Monitor KPIs.
 - ☐ Conduct stakeholder reviews.
 - ☐ Capture lessons learned.
 - ☐ Publish executive dashboards.
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Templates

RAID Log

Category	Details
Risks	
Assumptions	
Issues	
Dependencies	

Program Dashboard

KPI	Status
Schedule	Green
Budget	Green
Risks	Yellow
Adoption	Green
Quality	Green

Stakeholder Matrix

Stakeholder	Influence	Engagement
CTO	High	Weekly
VP Engineering	High	Weekly
Product Teams	Medium	Bi-Weekly
Operations	Medium	Monthly

Common Mistakes

1. Treating AI initiatives as standalone projects.
2. Ignoring governance requirements.
3. Failing to maintain RAID logs.
4. Neglecting stakeholder communication.
5. Underestimating change management.
6. Tracking outputs instead of outcomes.
7. Failing to establish measurable KPIs.

Key Takeaways

- AI TPMs are responsible for coordinating people, processes, and technology across AI programs.
 - Successful AI programs require strong governance, communication, and risk management.
 - RAID logs, roadmaps, and dashboards are essential tools for program management.
 - Azure DevOps and Power BI provide visibility into delivery performance.
 - Executive sponsorship significantly improves the likelihood of success.
 - Program Management is ultimately about delivering measurable business outcomes.
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Footer
